演1

$S_1 \xrightarrow{v_1} S_2 \xrightarrow{v_2} \phi$

1.

$v_1 = k_1[S_1]$

$v_2 = k_2[S_2]$

2.

$$\frac{d[S_2]}{dt} = v_1 - v_2$$

$$= k_1[S_1] - k_2[S_2]$$

演2

1.

$$\frac{d[Glu]}{dt} = -k[Glu]$$

$$\frac{d[Glu]}{[Glu]} = -k\,dt$$

$$\int \frac{d[Glu]}{[Glu]} = \int -k\,dt$$

$\log[Glu] = -kt + C$ (C:積分定数)

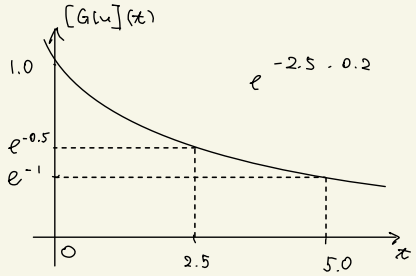
$[Glu] = e^{(-kt+C)}$

$= e^C \cdot e^{-kt}$

$t=0$ のとき $[Glu] = 1.0\,mM$ より、 $Glu = 1.0 \cdot e^{-kt}$

$= e^{-kt}$

2.



$e^{-0.2 \cdot 5}$

$e^{-2.5 \cdot 0.2}$

e^{-1}

演習3

```
import matplotlib.pyplot as plt

glu = 1.0
t = 0.0
delta_t = 0.01
endtime = 5.0

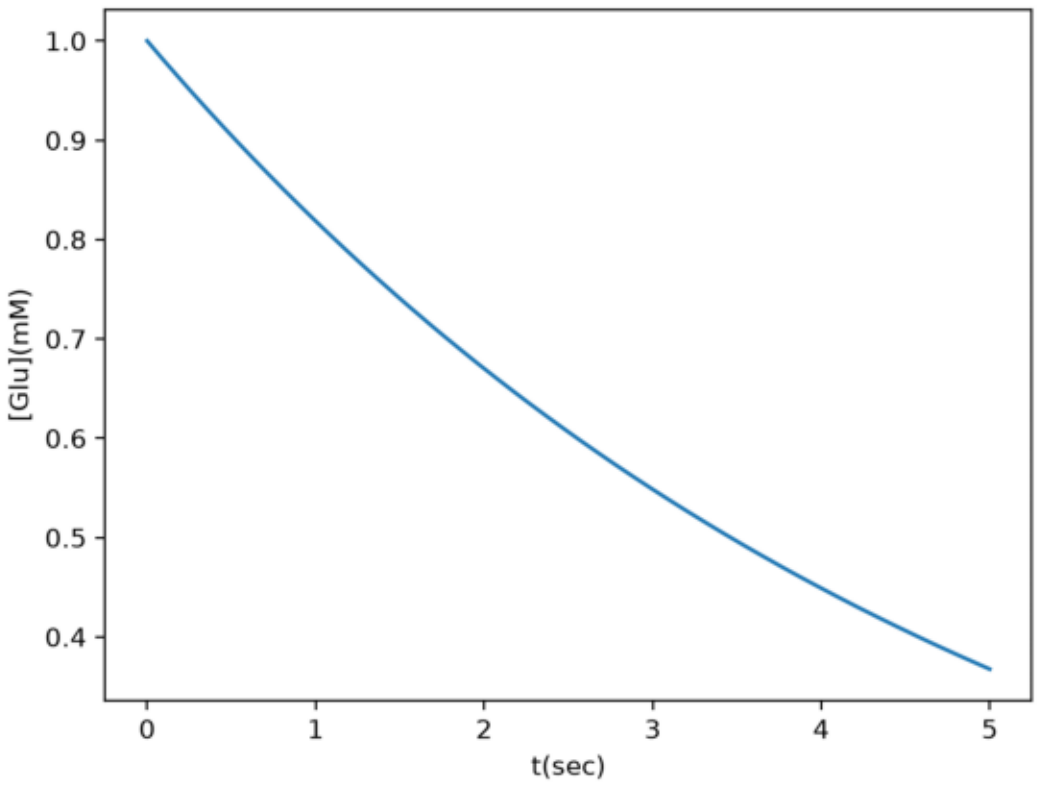
k = 0.2

glu_list = [ glu ]
t_list = [ t ]

while t < endtime - delta_t / 2 :
    dgludt = - k * glu
    delta_glu = dgludt * delta_t
    glu = glu + delta_glu
    t = t + delta_t

    glu_list.append( glu )
    t_list.append( t )

plt.plot( t_list , glu_list )
plt.xlabel( "t(sec)" )
plt.ylabel( "[Glu](mM)" )
plt.show()
```



演習4

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.integrate import solve_ivp

def f( t , s0 , Vmax1 , Km1 , Vmax2 , Km2 ):
    S1 = s0[ 0 ]
    S2 = s0[ 1 ]

    v1 = Vmax1 * S1 / ( Km1 + S1 )
    v2 = Vmax2 * S2 / ( Km2 + S2 )

    ds1dt = - v1
    ds2dt = v1 - v2
    return [ ds1dt , ds2dt ]

s0 = [ 2.0E-3 , 0 ]
t_span = [0, 30]
t_eval = np.linspace( t_span[0] , t_span[1] , 100 )

Vmax1 = 1.0E-3
Km1 = 1.0E-3

Vmax2 = 1.0E-4
Km2 = 2.0E-4

solution = solve_ivp( f, t_span, s0 ,
                      t_eval=t_eval , method='RK45',
                      args=( Vmax1 , Km1 , Vmax2 , Km2 ) )

plt.plot( solution.t , solution.y[0] )
plt.plot( solution.t , solution.y[1] )
plt.show()
```

